2.6 Quiz: Trigonometry — Markscheme

Version (a)

Total: 32 marks. Accuracy marks need 3 s.f., the exact value, or the calculator display with an ellipsis. In Problem 3 the calculator must be in radian mode: a value worked in degree mode earns the M and substitution marks if the method is right, and A0 for the answer.

1. [Maximum mark: 7]

(a) $\tan 32^\circ = \dfrac{FT}{35}$ **M1**

$FT = 35\tan 32^\circ = 21.9\,\text{m} \quad (21.8704\ldots)$ **A1**

Wrong methods: $35\sin 32^\circ = 18.5$ *or* $\frac{35}{\tan 32^\circ} = 56.0$ *earn M0A0.*

(b) $AT = \dfrac{35}{\cos 32^\circ}$ **M1**

$AT = 41.3\,\text{m} \quad (41.2712\ldots)$ **A1**

Accept Pythagoras with 35 *and* FT*, or* $\frac{FT}{\sin 32^\circ}$*. Wrong method:* $\frac{35}{\sin 32^\circ} = 66.0$ *earns M0A0. FT from (a).*

(c) $BF = 35 - 15 = 20$ **A1**

$\tan\theta = \dfrac{21.8704\ldots}{20}$ **M1**

$\theta = 47.6^\circ \quad (47.5578\ldots)$ **A1**

M1 for $\tan\theta = \frac{height}{their\ BF}$*. Wrong method: using* 15 *as the horizontal distance gives* 55.6°*: A0M1A0. FT from (a).*

2. [Maximum mark: 8]

(a) $AB^2 = 12^2 + 8^2 - 2(12)(8)\cos 105°$ **M1A1**

$AB^2 = 257.693\ldots$

$AB = 16.1\,\text{km}$ $\quad (16.0528\ldots)$ **A1**

M1 for the cosine rule, A1 for the correct substitution. Wrong method: adding $2(12)(8)\cos 105°$ *gives* 12.6*: M1A0A0.*

(b) $\dfrac{\sin H\hat{A}B}{8} = \dfrac{\sin 105°}{16.0528\ldots}$ **M1A1**

$H\hat{A}B = 28.8°$ $\quad (28.7751\ldots)$ **A1**

M1 for the sine rule, A1 for pairing $HB = 8$ *with* $H\hat{A}B$*. Wrong method: pairing* 12 *with* $H\hat{A}B$ *gives* $46.2°$*, which is* $H\hat{B}A$*: M1A0A0. Accept the cosine rule. Accept* 28.7 *from the rounded* $AB = 16.1$*. FT from (a).*

(c) Area $= \frac{1}{2}(12)(8)\sin 105°$ **M1**

Area $= 46.4\,\text{km}^2$ $\quad (46.3644\ldots)$ **A1**

Accept any two sides with the angle between them. Wrong method: $\frac{1}{2}(12)(8) = 48$ *earns M0A0.*

3. [Maximum mark: 8]

(a) $BC^2 = 6^2 + 10^2 - 2(6)(10)\cos 1.1$ **M1A1**

$BC^2 = 81.5684\ldots$

$BC = 9.03\,\text{cm}$ $\quad (9.03152\ldots)$ **A1**

Degree mode gives $BC = 4.00$*: M1A1A0.*

(b) Area $= \frac{1}{2}(6)(10)\sin 1.1$ **M1**

Area $= 26.7\,\text{cm}^2$ $\quad (26.7362\ldots)$ **A1**

Degree mode gives 0.576*: M1A0.*

(c) $\dfrac{\sin A\hat{C}B}{6} = \dfrac{\sin 1.1}{9.03152\ldots}$ **M1A1**

$A\hat{C}B = 0.634$ $\quad (0.633617\ldots)$ **A1**

M1 for the sine rule, A1 for pairing $AB = 6$ *with* $A\hat{C}B$*. Wrong method: pairing* 10 *with* $A\hat{C}B$ *gives* 1.41*, which is* $A\hat{B}C$*: M1A0A0. The answer must be in radians:* $36.3°$ *earns M1A1A0. Accept the cosine rule. FT from (a).*

4. [Maximum mark: 9]

(a) $A\hat{C}B = 180° - 68° - 47° = 65°$ **A1**

(b) $\dfrac{AC}{\sin 47°} = \dfrac{40}{\sin 65°}$ **M1A1**

$AC = 32.3\,\text{m}$ $\quad (32.2783\ldots)$ **A1**

M1 for the sine rule, A1 for pairing AC with $47°$ *and* 40 *with* $65°$*. Wrong method: pairing AC with* $68°$ *gives* 40.9*, which is BC: M1A0A0. FT from (a).*

(c) $\text{Area} = \frac{1}{2}(40)(32.2783\ldots)\sin 68°$ **M1**

$\text{Area} = 599\,\text{m}^2$ $\quad (598.559\ldots)$ **A1**

M1 for two sides with the angle between them. Wrong method: using $\sin 65°$ *with AB and AC gives* 585*: M0A0. FT from (b).*

(d) $\frac{1}{2}(24)(35)\sin\theta = 320$ **M1**

$\sin\theta = \frac{16}{21}$, $\quad$ acute value $49.6°$ **(A1)**

$\theta = 180° - 49.6324\ldots° = 130°$ $\quad (130.367\ldots)$ **A1**

Giving only the acute value $49.6°$ *earns M1A1A0. A1 for* 130 *(3 s.f.) or the display with an ellipsis;* 130.4 *with no ellipsis earns A0.*